A Bounded Perturbation Which Spoils a Heat-Trace Expansion

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2026-07-18

Source Signal

Eckstein–Zajac, arXiv:1412.5100, study asymptotic and exact expansions of heat traces for positive unbounded operators with compact inverses. In their concluding discussion, p. 38, they point to the behavior of the heat trace under bounded perturbations as a natural next step: a bounded perturbation does not change the leading small-time behavior, but may spoil the full asymptotic expansion.

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In Section 4 devoted to examples we always worked with operators, the spectrum of which is known explicitly. Therefore, the operatorial aspect of the problem was somewhat hidden. In practice, one is rarely granted the comfort of knowing the full spectrum of a given operator. Even if this is the case, one would like study the behaviour of the heat trace when a fixed operator P is perturbed to $P + A$, with some bounded A. Clearly, a bounded perturbation of P would not change the leading behaviour of $\mathrm{htr}_P(t)$ at small t, but it can, at least in principle, spoil the asymptotic expansion.

Indeed, perturbations may drastically change the analytic properties of associated zeta-functions. For instance, the modulus of the Dirac operator on the standard Podleś sphere has (up to a multiplicative constant) the following eigenvalues $\lambda_n(|\mathcal{D}_q|) = q^{-n} - q^n$ (see [26]). It can thus be considered a sum of an operator of exponential spectrum P and a trace class perturbation Q. It turns out that the poles of ζ_{P+Q} form a regular lattice on the left complex half-plane [28], whereas the poles of ζ_P are located only on the imaginary axis (see Section 4.2). Although, the convergence properties of the small t expansion of $\mathrm{htr}_{P+Q}(t)$ are not altered by the perturbation Q, but the estimates of contour integrals are much more subtle and tedious to control (see [28, Proposition 4.3]).

We regard the investigation of the impact of perturbations on the asymptotic expansion of heat traces to be an important and natural next step in the study initiated in this paper. We hope that a combination of our results with the techniques developed in [7, 53, 65] can lead to a better understanding of heat traces outside the realm of classical pseudodifferential

Result

Theorem 1. *There are a positive self-adjoint compact-resolvent operator P_0 on $\ell_2(\mathbb{N})$ and a bounded positive self-adjoint diagonal operator A with $0 \leq A \leq I$ such that:*

1. $\mathrm{Tr}(e^{-tP_0})$ *has an exact Laurent expansion at $t = 0$;*
2. $P = P_0 + A$ *still satisfies* $\mathrm{Tr}(e^{-tP}) \sim t^{-1}$ *as $t \downarrow 0$;*

3. there is no constant c such that

$$\mathrm{Tr}(e^{-tP}) = t^{-1} + c + o(1).$$

Thus an arbitrary bounded perturbation can destroy the heat-trace expansion already at the constant term.

Proof Intuition

For $P_0 e_n = n e_n$, the heat trace is the geometric series $(e^t - 1)^{-1}$. A diagonal 0/1 perturbation changes only selected eigenvalues, replacing e^{-tn} by $e^{-t(n+1)}$ on a set $S \subset \mathbb{N}$. The error is therefore $(e^{-t} - 1)\sum_{n\in S} e^{-tn}$. If S has no Abel density, this error has no limit at order t^0.

Proof

Let $H = \ell_2(\mathbb{N})$ with standard unit vector basis $(e_n)_{n\geq 1}$ and set

$$P_0 e_n = n e_n.$$

Then P_0 is positive, self-adjoint, and has compact inverse. Its heat trace is

$$H_0(t) = \mathrm{Tr}(e^{-tP_0}) = \sum_{n=1}^{\infty} e^{-tn} = \frac{1}{e^t - 1} = t^{-1} - \frac{1}{2} + O(t).$$

Choose integers $N_1 < N_2 < \cdots$ such that $N_{k+1}/N_k \to \infty$; for example, $N_1 = 2$ and $N_{k+1} = N_k^2$. Put

$$S = \bigcup_{m\geq 1} \{N_{2m}, N_{2m} + 1, \ldots, N_{2m+1} - 1\}.$$

Define the bounded diagonal operator A by

$$A e_n = \mathbf{1}_S(n) e_n.$$

Then $0 \leq A \leq I$, and $P = P_0 + A$ is again positive self-adjoint with compact inverse. Write

$$F(t) = \sum_{n\in S} e^{-tn}.$$

Since the spectra are diagonal,

$$H(t) := \mathrm{Tr}(e^{-tP}) = \sum_{n\notin S} e^{-tn} + \sum_{n\in S} e^{-t(n+1)} = H_0(t) + (e^{-t} - 1)F(t).$$

Because $0 \leq F(t) \leq H_0(t)$, the perturbation term is $O(1)$, hence $tH(t) \to 1$.

We now show that the constant term cannot exist. Let $t_m = N_{2m+1}^{-1}$. The contribution of the occupied block $[N_{2m}, N_{2m+1})$ gives a Riemann sum:

$$t_m \sum_{n=N_{2m}}^{N_{2m+1}-1} e^{-t_m n} \longrightarrow \int_0^1 e^{-x}\, dx = 1 - e^{-1}.$$

The earlier blocks contribute at most $t_m N_{2m-1} \to 0$, and the later blocks contribute at most a geometric tail beginning at N_{2m+2}, which is $o(1)$ after multiplication by t_m. Therefore

$$t_m F(t_m) \longrightarrow 1 - e^{-1}.$$

Next choose $M_m = \lfloor (N_{2m+1} N_{2m+2})^{1/2} \rfloor$ and $u_m = M_m^{-1}$. The point M_m lies deep in the empty gap after N_{2m+1}. All earlier occupied blocks contribute at most $u_m N_{2m+1} \to 0$, while all later occupied blocks begin after N_{2m+2} and contribute $o(1)$ after multiplication by u_m. Hence

$$u_m F(u_m) \longrightarrow 0.$$

Finally, since $e^{-t} - 1 = -t + O(t^2)$ and $F(t) \le H_0(t) = O(t^{-1})$,

$$(e^{-t} - 1)F(t) = -tF(t) + o(1).$$

Thus

$$H(t_m) - t_m^{-1} \longrightarrow -\frac{1}{2} - (1 - e^{-1}),$$

whereas

$$H(u_m) - u_m^{-1} \longrightarrow -\frac{1}{2}.$$

The quantity $H(t) - t^{-1}$ has two different subsequential limits as $t \downarrow 0$. Consequently no constant c can satisfy $H(t) = t^{-1} + c + o(1)$, and the heat trace of $P_0 + A$ has no classical asymptotic expansion beyond the leading term.

Verification And Scope

The perturbation is bounded, positive, self-adjoint, and diagonal, but it is not compact. The result therefore addresses arbitrary bounded perturbations of compact-resolvent Hilbert-space operators. It does not contradict structured preservation theorems for elliptic, pseudodifferential, trace-class, or regular spectral-triple perturbations.

Local index search found no exact prior row for arXiv:1412.5100. Web searches for bounded perturbations of heat-trace asymptotics found preservation statements in structured settings but no exact diagonal counterexample of the form above.

References

- M. Eckstein and A. Zajac, "Asymptotic and exact expansions of heat traces", arXiv:1412.5100; Math. Phys. Anal. Geom. 18 (2015), article 28.